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Antifouling behavior of self-renewal acrylate boron polymers with pyridine-diphenylborane side chains†

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Currently, most high-performance tributyltin (TBT)-free antifouling paints contain cuprous oxide (Cu_2O) in their formulations, leading to a serious challenge to the marine environment and ecosystem. In this study, non-metal acrylate boron polymers (ABPs) were synthesized using acrylate acid polymers (AAPs) as the backbone and a pyridine-phenyl borane compound as the side chain with hydrolyzable functional groups. The antifouling mechanism of ABPs researched by XPS and EDS indicated that the antifouling performance of ABPs was realized by the hydrolysis of hydrophobic diphenyl borane pyridine side groups. The results of diatom anti-setting test showed that fewer or no algae cells were attached on the surfaces of ABPs with higher contents of hydrophobic diphenyl borane pyridine side groups. The growth rates of algae suspension solutions containing ABPs were close to that of the blank solution without ABPs. Antifouling paints made of ABPs showed excellent antifouling performances in the Yellow Sea of China. This study provides new insights into the development of ABPs with desirable antifouling properties and environmental friendliness.

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1. Introduction

Artificial surfaces immersed in seawater are subjected to the attachment of thousands of marine organisms such as bacteria, diatoms and molluscs, which is known as marine biofouling.¹ Organisms attached to ships, offshore oil platforms, heat exchangers, oceanographic sensors, aquaculture systems and submarine pipelines have plagued the society for many years.^{2–6} This unwanted colonization not only increases the fuel consumption, navigating drag and acceleration of corrosion but also blocks the valves and pipes of seawater intake installations and hampers the accuracy and the reliability of oceanographic sensors, initiating a series of problems.⁷ In the middle of 1960s, a class of self-polishing paints containing tributyltin (TBT)-based compounds with excellent antifouling properties and modest costs of production was reported. However, the International Maritime Organization (IMO) banned their use because of the damage they inflicted on the marine ecological environment.⁸ Currently, copper-based antifouling paints are used as substitutes

for TBT paints, but high concentrations of copper can also be harmful to marine organisms.⁹ Therefore, in the past few decades, eco-friendly antifouling paints without heavy metal ions have attracted the interest of many researchers, *e.g.*, low surface-energy fouling-release coatings,^{10–12} zwitterionic polymers,^{13,14} and amphiphilic polymers.^{15–18} However, currently, there is no mature or efficient antifouling method that can completely replace the self-polishing antifouling technology.

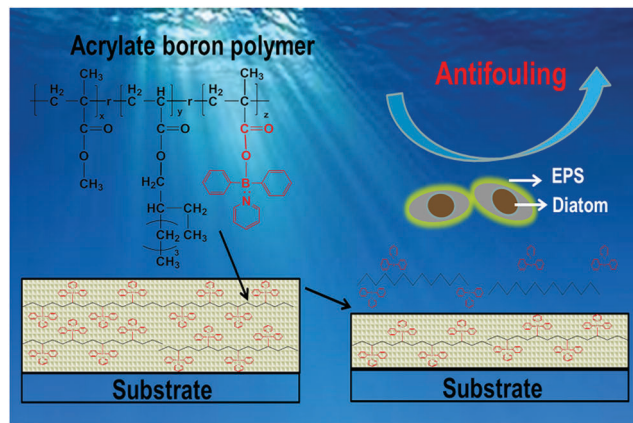
Typical copper-free self-polishing antifouling coatings are based on traditional heavy metal-containing polymers such as zinc, copper, and trialkyltitanate-based (meth)acrylate copolymers, which release organometallic compounds during the self-renewal process.^{19–21} Another self-polishing non-metal substitute alkylsilyl (meth)acrylate-based polymers are well-known for their sustained stable hydrolysis rate and good antifouling performance.^{22,23} However, the self-polishing rate of the alkylsilyl (meth)acrylate-based polymer is very low to obtain good antifouling performance in a static marine environment when the alkylsilyl (meth)acrylate content is below 45%.²⁴ A class of compounds having a boron atom attached to at least one $\text{C}_8\text{--C}_{26}$ fatty acid residue possessing at least one $\text{C}=\text{C}$ moiety was reported by William Brenden Carlson.²⁵ Polymers polymerized by these compounds exhibit antifouling properties when used as coatings. However, these organic boron polymers are synthesized using complicated procedures over a period of couple of days, and the hull of a boat should be preheated to a suitable temperature to cure the

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Scheme 1 Antifouling process of ABPs.

polymer. The complicated synthetic methods and construction methods limit their application in the antifouling field.

Herein, we proposed a simple method to synthesize non-metal ABPs with self-renewal property by the protonolysis reaction of AAPs and pyridine-triphenylborane (PTPB). A set of ABPs was acquired by varying the monomer ratio of MAA. To investigate the antifouling and eco-friendly properties of the developed ABPs, the weight loss measurement, diatom settlement, diatom growth rate assays and antifouling paint raft test were carried out. Up to now, there is no comprehensive literature review to explain the antifouling mechanism of ABPs. In this study, we investigated the antifouling mechanism of ABPs by EDS and XPS, and the results demonstrated that the antifouling performance of ABPs was closely related to the hydrophobic diphenyl borane pyridine side groups. The diatom anti-setting assays and the antifouling coatings' raft test showed that ABPs exhibited strong resistance to fouling. This study will hopefully provide an eco-friendly way for the design of new effective antifouling coatings (Scheme 1).

2. Experimental

2.1 Materials

Tetrahydrofuran (THF), xylene, 1-methoxy-2-propanol (PGM), toluene, absolute ethanol, butyl alcohol, methyl methacrylate (MMA), 2-ethylhexyl acrylate (2-EHA), methacrylic acid (MAA), azobisisobutyronitrile (AIBN), and azobis-2-methylbutyronitrile (AMBN) were purchased from the Tianjin Guangfu Fine Chemical Research Institute (Tianjin, China). PTPB was purchased from Changzhou Kean Chemical Technology Co., Ltd. *tert*-Butyl peroxy-2-ethylhexanoate (OT) was purchased from HWRK Chem Co., Ltd (Beijing, China). 2,4-Diphenyl-4-methyl-1-pentene (MSD) was purchased from Adamas Reagent Co., Ltd (Shanghai, China). ZnPT (zinc pyrithione) was purchased from Wuxi Zhufeng Fine Chemical Co., Ltd (Jiangsu, China). Deionized water was purified using a Milli-Q Plus system (Millipore, Schwalbach, Germany). Artificial seawater (ASW) used for diatom cultivation was prepared according to ASTM

D1141-98, whereas ASW in the film immersion experiment was prepared using reef salt provided by Seachem. All reagents were used as received without further purification.

2.2 Synthesis of the boron polymer

As shown in Fig. 1, ABPs were synthesized using a two-step method. First, MMA, 2-EHA, and MAA were used to synthesize acrylate acid polymers (AAPs) *via* radical copolymerization initiated by AIBN, AMBN and OT; then, ABPs were prepared from AAPs and PTPB.

2.2.1 Synthesis of AAPs. AAPs were synthesized *via* a facile free radical polymerization. First, 106.6 parts of xylene and 33.3 parts of butyl alcohol were added to a four-necked flask equipped with a cooling tube, a thermometer, a dropping funnel and a stirrer. Then, the mixed solvent was heated to 95–100 °C. Next, a mixture of 30.9–39.1 parts of MMA, 44.1–55.9 parts of 2-EHA, 5–25 parts of MAA, 1 part of MSD, 2.0 parts of AIBN, and 5.6 parts of AMBN was added dropwise. After the completion of the addition, 20.0 parts of xylene and 0.5 parts of OT were fed dropwise into the flask over 30 minutes at a constant speed and then, the solution was allowed to be continuously stirred for 1.5 h. Also, 6.7 parts of xylene were added to obtain the transparent AAPs. Then, the AAPs were isolated by precipitating into excess ice ethyl alcohol to remove the remaining unreacted monomers. Finally, after the addition of solvents, the polymers with a solid content of 39.6% were obtained.

¹H NMR (500 MHz, CDCl₃, ppm): 10.63 (CH₂CH(COOH)), 2.37 (CH₂CH(COOH)), 1.83 (CH₂CH(COOH)), 1.60–1.77 (CH₂C(COOCH₃)(CH₃)), 3.64 (CH₂C(COOCH₃)(CH₃)), 1.27 (CH₂C(COOCH₃)(CH₃)), 1.89 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)), 2.13 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)), 4.24–4.29 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)), 1.89 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)), 1.25–1.55 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)), 0.88–0.99 (CH₂CH(COOCH₂CH(CH₂CH₃)CH₂CH₂CH₂CH₃)). IR (cm⁻¹): 3250 (OH), 1740 (C=O), 990, 1160 (C–O–C).

2.2.2 Synthesis of ABPs. Typical reaction conditions were as follows: 6.5–33.5 parts of PTPB (excessive amount), 75 parts of xylene and 50 parts of THF were added to a 4-necked flask equipped with an agitator, a reflux condenser and a thermometer. The mixture was heated to 95–100 °C and then, 100 parts

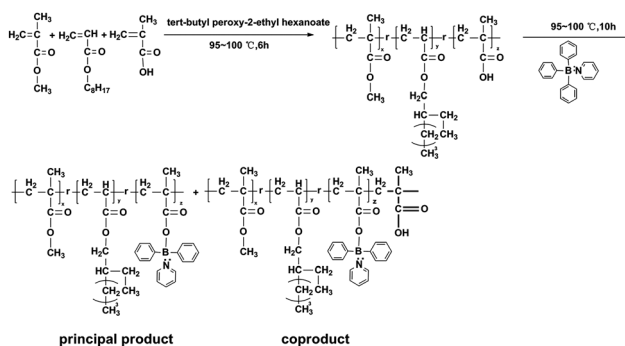


Fig. 1 Synthetic route for ABPs.

Table 1 Acid values and recipes for the synthesis of acrylate boron polymers (ABPs)

Sample	MMA/2-EHA/ MAA/PTPB ^a	Acid value (mgKOH g ⁻¹)	M _w (g mol ⁻¹)
ABP-1	30.9/44.1/25.0/33.5	95.27	1.1 × 10 ⁴
ABP-2	32.9/47.1/20.0/26.8	84.87	8.5 × 10 ³
ABP-3	35.0/50.0/15.0/19.9	70.58	8.3 × 10 ³
ABP-4	37.1/52.9/10.0/13.1	49.76	6.9 × 10 ³
ABP-5	39.1/55.9/5.0/6.5	24.49	5.8 × 10 ³

^a Feed weight ratio.

of AAP were added into the mixture. Furthermore, the mixture was stirred for 10 h and it turned from a milky white and cloudy solution to a light brown and transparent system. A vacuum filtration system was used to remove the unreacted PTPB and other impurities. To ensure a constant solid content for determining the viscosity of the polymers and endow the polymers with proper viscosity for different applications, the non-volatile contents of the polymers were adjusted to 40% through reduced pressure distillation and solvent addition. The chemical compositions of different reaction mixtures, acid values of ABPs and the sample abbreviations used in this study are summarized in Table 1. *M_w* and PDI of ABPs, as determined by gel permeation chromatography (GPC), were in the ranges from 5.7 × 10³ to 1.1 × 10⁴ g mol⁻¹ and from 2.0 to 3.4, respectively. Because of excessive PTPB, only a small part of carboxyl groups remained unreacted for steric hindrance. The degree of polymerization of MMA, 2-EHA and MAA and PTPB grafting ratio were determined by ¹H NMR and FTIR, respectively, as listed in Table 2. ¹H NMR (500 MHz, CDCl₃, ppm): 7.35–7.75 (B(C₆H₅)), 7.36–8.98 (B(C₅H₅N)). IR (cm⁻¹): 3250 (OH), 1740 (C=O), 1340 (B–O–C) (Fig. 2).

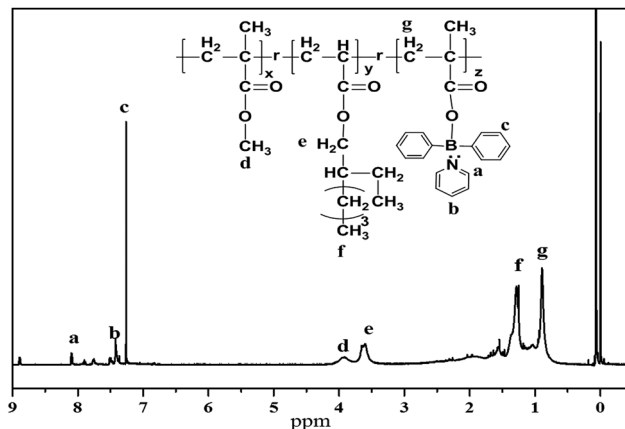
As we can see in Table 2, the degree of polymerization of MAA is higher than those of MMA and 2-EHA; the reason for this phenomenon may be the higher rate of polymerization under the reaction condition used. Also, the high degree of polymerization of MAA is beneficial to obtain high self-polishing rates and antifouling property. Table 2 also shows that most PTPB were grafted to AAPs, and a small amount of residual PTPB remained unreacted for steric effect.

2.3 Characterization

Fourier transform infrared (FTIR) spectroscopy measurements were recorded on a Perkin Elmer Spectrum 100 Fourier transform infrared spectrometer to investigate the molecular structures of acrylate acid polymers and ABPs in the region of 400–4000 cm⁻¹.

Table 2 Degree of polymerization of MMA, 2-EHA and MAA and PTPB grafting ratios during the protonolysis reaction of AAPs and pyridine-triphenylborane

Sample	DP _{MMA}	DP _{2-EHA}	DP _{MAA}	PTPB grafting ratio (%)
ABP-1	5	11	25	100
ABP-2	4	9	19	96
ABP-3	4	9	19	92
ABP-4	3	8	15	95
ABP-5	3	6	13	87

**Fig. 2** ¹H NMR spectrum of ABPs.

The molecular weights and molecular weight distributions (*M_w* and PDI) of AAPs and ABPs were measured on a waters 1515 gel permeation chromatography (GPC) system (with THF as the eluent at a flow rate of 0.4 mL min⁻¹ at 30 °C). Polystyrene (PS) standards (400–43 000 g mol⁻¹) were used to calibrate the instrument. ¹H NMR (500 MHz) measurements were obtained on a Bruker Advance III-500 spectrometer with deuterated chloroform (CDCl₃) as the solvent at room temperature. Thermogravimetric analysis (TGA) was performed on a TA Q50 apparatus under nitrogen atmosphere at a heating rate of 20 °C min⁻¹ in the range of 30–500 °C. The thermogravimetric weight loss curve (TG, %) and the weight loss derivative curve (DTG, % °C⁻¹) were recorded as a function of temperature. Differential scanning calorimetry (DSC) measurements were performed on a TA Q20 apparatus between –20 and 100 °C at a rate of 20 °C min⁻¹ with a sample weight of 5–10 mg. The viscosities of AAPs and ABPs were measured on a viscosimeter (DV-C, Brookfield, USA). The water contact angle (CA) measurements on surfaces were carried out by a video optical contact angle system (Edmund Optics, Barrington, NJ, USA) at room temperature. For each surface sample, the reported result was the average of at least five measurements and the error bars represent the standard deviation. The acid values (AV) were found by titration to determine the hydrolysable groups in AAPs and ABPs. The titrant used was a potassium hydroxide (KOH) solution at a concentration of 0.1 mol L⁻¹. Before titration, a mixed solvent was prepared by combining 95% ethanol (v/v) solution with toluene at a volume ratio of 1:2. Then, approximately 0.5 g of the test polymer was added to an Erlenmeyer flask. Afterwards, 50 mL of the mixed solvent was used to dissolve the polymer. Bromothymol blue in ethanol solution was then added to the polymer solution. During the titration, potassium hydroxide solution was added dropwise to the polymer solution containing the indicator. When the titration end point was reached, the colour of the solution turned from yellow to blue. The acid value (mgKOH g⁻¹) can be calculated by the following eqn (1):

$$AV = \frac{56.1 \times V \times 0.1}{M \times NV} \quad (1)$$

Here, AV, M and NV are the acid value, the weight and the solid content of the test polymer, respectively, and V is the volume of KOH used in the titration.

2.4 XPS and SEM

The microtopography of ABPs was evaluated by a scanning electron microscope (SEM) equipped with a spectrometer for energy-dispersive X-ray spectroscopy (EDS) (JEOL JSM-6480A microscope). The chemical states of the polymers' surfaces were evaluated by X-ray photoelectron analysis on a multi-functional X-ray photoelectron spectroscopy (XPS) instrument (PHI-5700, Perkin-Elmer, USA).

2.5 Weight loss measurement

Measured samples were placed in a square aquarium into which artificial seawater (ASW) was added and changed every week. The aquarium was kept at 21.6 ± 1 °C. Before the experiment, glass slides without the top-coated polymer were weighed (W_{panel}), and the weight of each dried coated glass slide was recorded as W_0 . After different time periods, the glass slides were removed from ASW and immersed in distilled water for 4 h. Then, all these glass slides were put into an oven and dried at 100 °C for 6 h before their weights ($W_{t,\text{dry}}$) were recorded. The weight difference between W_0 and $W_{t,\text{dry}}$ shows the weight loss of the polymer from each coating. Each polymer was smeared on three glass slides for testing. Therefore, the used and reported weight loss ratios represent the average from three identical coatings. The weight loss ratio (WLR) can be calculated by the following eqn (2):

$$\text{WLR} = \frac{W_0 - W_{t,\text{dry}}}{W_0 - W_{\text{panel}}} \times 100\% \quad (2)$$

Here, $W_{t,\text{dry}}$, W_0 , and W_{panel} are the weights of the dried coated panel at time t , the coated panel at time $t = 0$ and the initial non-coated panel, respectively. Error bars represent the standard deviation of data.

2.6 Diatom settlement assay and growth inhibition test

The algae *Nitzschia closterium* were used as a model organism to evaluate the antifouling performance of ABPs. A stock solution of *Nitzschia closterium* from the Center for Collections of Marine Algae of Xiamen University (CCMA), (Pearl River Estuary, China) was grown and cultured in F/2 medium without aeration at 21 ± 2 °C with a 12/12 h light:dark (L/D) cycle of fluorescent illumination at 2000 lux, and it was gently stirred twice daily. The test ABPs and the control polymer (acrylate zinc self-polishing polymer) were coated on glass slides. All the samples and a pure glass slide were placed on the bottom of the aquarium with a suspension solution containing at least 1×10^5 algal cells per mL. The aquarium was placed in an illumination incubator. The parameters of the incubator were set as described above. The samples were incubated for 10 days under the above conditions and then gently lifted out of the aquarium. All the substrates were washed with deionized water to remove any unsettled diatoms. An optical microscope (Leica DML 300B, Germany) was used to

observe and record the morphology of the diatoms attached on the substrate with a confidence value of $p < 0.05$.

The growth inhibition test was carried out as follows: after the algae cell number reached 1×10^5 , 100 mL of the algae solution was added into individual conical flasks containing a test glass slide and prepared as described before. The micro-algae cell concentration was counted on a cell count chamber haemocytometer with an optical microscope (Leica DML 300B, Germany) and correlated with the optical densities of the microalgae samples at 314.5 nm based on linear fitting equations. During the incubation period, the concentration of the algae suspension solution was quantified on a UV-2550-Visible Spectrophotometer (Shimadzu Corporation, Japan). The growth rate of algae was calculated by the following eqn (3):

$$\mu = t^{-1} \cdot \ln(N_t/N_0) \quad (3)$$

Here, N_t , N_0 and t are the algae number at time t , algae number at time $t = 0$, and the number of test days. The results represent the average of three parallel tests, and the standard deviations are presented as error bars for quantitative comparison.²⁶ Statistical analysis was carried out in the same way as that for the diatom settlement assay.

2.7 Preparation of copper-free antifouling paints and raft test

Five acrylate boron polymer coatings (ABPC) with ABP-1, ABP-2, ABP-3, ABP-4, and ABP-5 as binders were formulated to carry out the raft test. Typically, 52.1 g ABP, 5 g ZnPT, 5 g tralopyril, 28.5 g pigments, 1.5 g chlorinated paraffin, 2.0 g anti-settling agent, and 5.9 g solvent were mixed uniformly with glass beads by a mechanical stirrer at a speed of 40 rpm for 30 minutes; then, the antifouling paints were filtered to remove the glass beads. After the filtration, the paints were packaged for use. Antifouling paints were sprayed on tinplate plates to fabricate antifouling coatings for further analysis.

The copper-free antifouling paints were coated on fibreglass-reinforced phenolic resin panels pre-coated with Jotun Tie Coat Penguard Midcoat M20 with a dry film thickness of 160 µm, and the coated plates were secured with insulated wire inside a stainless frame, leaving 8 mm gap between each plate and 30 mm gap between the plates and the top and the bottom of the frame. The frames (with eight plates, and the size of each plate was 300 mm × 100 mm) were deployed from a dock at Qingdao NCS Testing Protection Technology Co., Ltd (36°03'N, 120°25'E) on August 2nd, 2016. The frames were suspended at 1 m depth and sampled every 3 months to record the settlement and growth of fouling organisms.

3. Results and discussion

3.1. Synthesis of ABPs

The preparation of ABPs was carried out using a two-step procedure: (1) synthesis of acrylate acid polymers and (2) synthesis of ABPs. Because the absorption peaks of all ABPs and AAPs were almost the same, ABP-1 was chosen to compare with AAP-1 with regard to the most distinct peaks in Fig. 3(a).

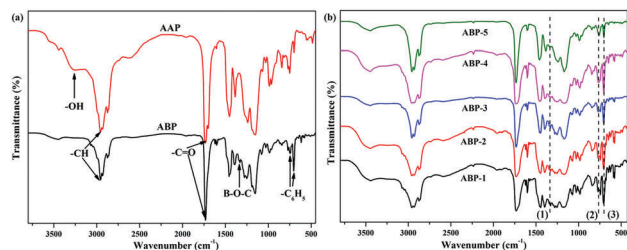


Fig. 3 IR spectra of AAP-1 and ABP-1 (a). IR spectra of ABP-1, ABP-2, ABP-3, ABP-4, and ABP-5 (b). (1) 1340 cm^{-1} , (2) 750 cm^{-1} , (3) 700 cm^{-1} .

The peaks at 3250–3350 cm^{-1} were ascribed to the stretching vibration of –OH groups in the carboxyl moieties of AAPs. The peak at 2940 cm^{-1} indicated the presence of –CH bonds. The peak at 1740 cm^{-1} corresponded to the stretching vibrations of the carbonyl bonds of the ester groups in AAPs and ABPs. By comparing the spectrum of AAPs with that of ABPs, we found a new peak at 1340 cm^{-1} , which was ascribed to B–O–C bonds.²⁷ In addition to the presence of a new peak, another difference could be found: the peaks for hydroxyl groups disappeared in the spectrum of ABPs (the PTPB grafting ratio can be seen in Table 2). When we increased the content of pyridine-triphenylborane, the intensities of the peak at 1340 cm^{-1} corresponding to the B–O–C bonds and those of the peaks at 700 cm^{-1} and 750 cm^{-1} corresponding to the out-of-plane vibration of –CH bonds in mono-substituted benzene increased at the same time. The peaks corresponded to the molecular structure described in patent CN101679787B written by PPG Co., Ltd.²⁸ The FTIR results were consistent with the results of ^1H NMR. All these results demonstrated that ABPs were successfully synthesized (Fig. 3).

3.2. Diatom inhibition test of different ABPs

To evaluate the environmentally friendly properties of ABPs, the diatom inhibition test was performed to determine whether the antifouling effect is limited to the surface of the polymer or extended into the whole bulk solution. The cell number of algae in the solutions with blank glass slides was slightly higher than that of the others, but the growth rates of algae solutions were almost the same. According to the equation $\mu = t^{-1} \cdot \ln(N_t/N_0)$, the growth rate with ABP-1 was $1.86 \times 10^{-3} \text{ day}^{-1}$, the growth rate with ABP-3 was $1.97 \times 10^{-3} \text{ day}^{-1}$, the growth rate with ABP-5 was $2.03 \times 10^{-3} \text{ day}^{-1}$, and the growth rate with blank was $2.21 \times 10^{-3} \text{ day}^{-1}$. Therefore, ABPs are hypotoxic to nontarget fouling organisms in seawater when the binders are applied on the hull of a boat and show an antifouling effect when organisms are attached to the surface of the polymers. The test showed the environmentally friendly properties of ABPs (Fig. 4).²⁹

3.3. Diatom anti-setting test of different ABPs

The diatom anti-setting assays could examine the antifouling performance and hydrolysis properties of ABPs to some extent. The number of attached cells of the *Nitzschia closterium* diatom had a negative correlation with the acid value of binders. More algae cells were found on the surface of the blank glass slide

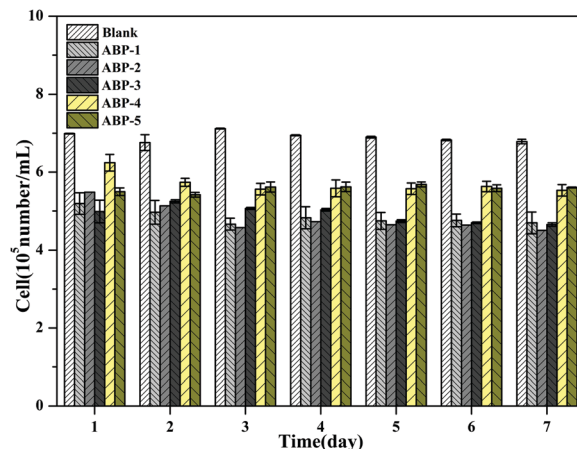


Fig. 4 The cell number of diatoms in solutions with various ABPs. Each point is the average of 180 fields of view counted out on six replicate slides. Error bars represent 95% confidence limits.

than on the surfaces of different kinds of polymers after two weeks in the suspension. To compare the antifouling properties of ABPs with those of an acrylate zinc polymer, the acrylate zinc polymer was coated on a glass slide and put into the algae solution for two weeks. The result showed that only a few algae cells were attached on the surface of the acrylate zinc polymer. The number of attached cells on the acrylate zinc polymer was similar to that on the surface of ABP-3, which indicated that the antifouling effect of the acrylate zinc polymer was somewhat similar to that of ABP-3 and better than that of ABP-4 and ABP-5 (Fig. 5).

ABPs with higher acid values also exhibited more labile esters (hydrophobic diphenyl borane pyridine side groups) and higher hydrolysis rates. When *Nitzschia closterium* attached to ABPs, the unstable surfaces were hydrolyzed and self-renewal was observed, which was not conducive to strong adhesion of diatoms; this led to less adhesion. To prove this hypothesis, tests (EDS and XPS) were carried out, and the results are described in the following text. The hydrolysis product diphenylhydroxyborane can also deter the diatom cells from attaching to the surface and then, the hypotoxic hydrolysis products diphenylhydroxyborane and phenyldihydroxyborane are degraded into micromolecular phenol and benzene by the sunlight and heavy metal ions, which is consistent with the results of numerous past experiments.²⁹

3.4. Antifouling paint raft test of different ABPs

To prove the practicability and antifouling performance of as-prepared ABP coatings, an ocean raft test was implemented in the Yellow Sea. The visual assessment of sample plates is shown in Fig. 6. As we can see from the digital images, after 3 months of immersion in natural seawater, no clear fouling was found on the plates coated with paints made of ABP-1, ABP-2, ABP-3, and ABP-4 except for some silt; meanwhile, slime was found on the surface of blank and on the coatings made of ABP-5. Also, more slime was attached to the blank compared with that to the coatings made of ABP-5. The reason for this phenomenon is

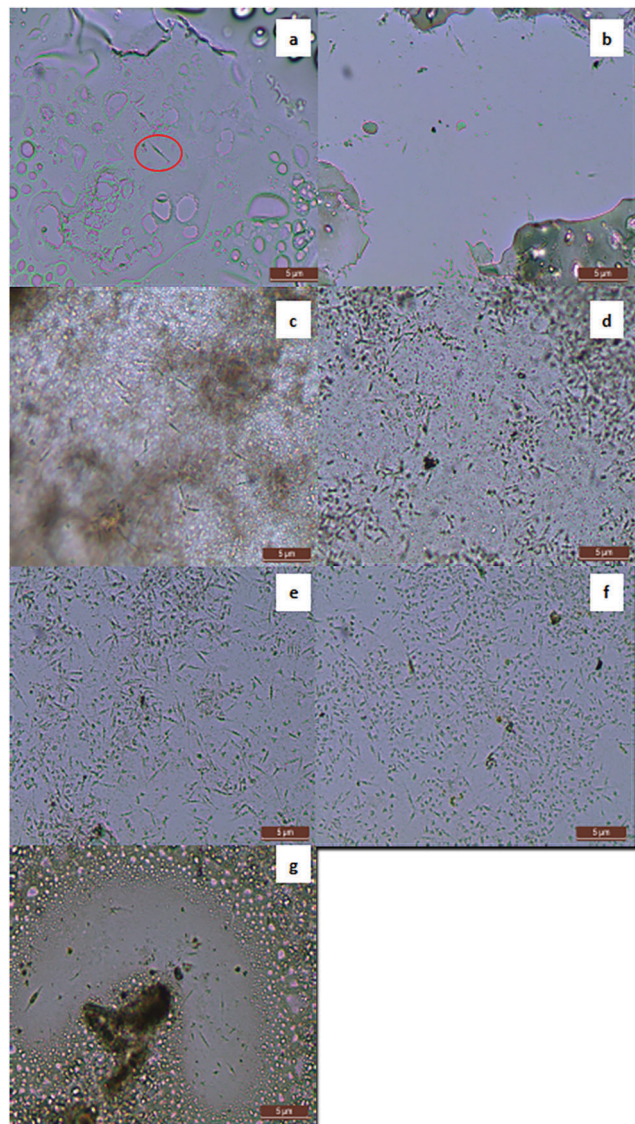


Fig. 5 Optical microscopy images of thin films (a) ABP-1, (b) ABP-2, (c) ABP-3, (d) ABP-4, (e) ABP-5 (f) blank glass slide (g) acrylate zinc polymer immersed in the algae solution for two weeks. The diatom in the circle is *Nitzschia closterium*.

that the antifouling paints made of ABPs are self-polishing antifouling paints; therefore, the antifouling agent release rate of the antifouling paints depends on the hydrolysis rate of ABPs. The ABPs with high hydrolysis rates have high antifouling agent release rates and thus have good antifouling performances. The results of the diatom anti-setting test of different ABPs showed that the order of hydrolysis rates of ABPs from high to low was ABP-1, ABP-2, ABP-3, ABP-4 and ABP-5. Thus, the antifouling efficacy sequence of the coatings from good to bad was also ABP-1, ABP-2, ABP-3, ABP-4 and ABP-5, and the antifouling efficacy of all the coatings was better than that of blank because no barnacles were found on other plates. After 12 months, some hard shell organisms adhered to the surface of the blank, whereas negligible fouling organisms were found on the surfaces of the copper-free antifouling paints made of ABP-1, ABP-2, ABP-3,

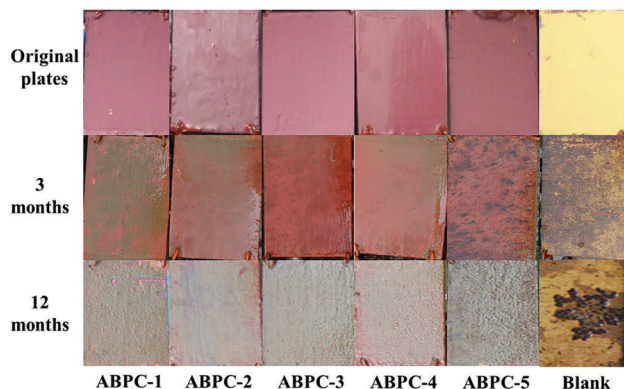


Fig. 6 Photographs of the ABPC coatings and blank immersed in the Yellow Sea of China for 3 months and 12 months.

and ABP-4 except some silt, which indicated that the copper-free antifouling paints containing ABPs have excellent antifouling properties during one year test.

3.5. Hypothesis of antifouling mechanisms of ABPs

We have then assumed that the antifouling properties of ABPs are associated with their surface morphologies and elemental distributions; the hydrophobic diphenyl borane pyridine side groups form hydrophobic domains on the film surface and the micro/nano hydrophobic raised domains can prevent the attachment of fouling organisms. Once the protective barrier is overcome by inorganic micromolecules and other impurities, fouling organisms begin to adhere to the surfaces. After the hydrophobic diphenyl borane pyridine side groups are peeled off from the molecular backbone through the self-polishing process, the polymers are hydrolysed, and the attached biofouling organisms peel off from the surface, resulting in an exfoliated coating surface; thus, the antifouling performance of ABPs is achieved by the hydrolysis of hydrophobic diphenyl borane pyridine side groups. To verify this assumption, a comparative study needs to be carried out to detect the surface morphology and elemental distribution.

3.6. Contact angles of ABPs

To check whether hydrolysable hydrophobic domains exist on polymer surfaces, the static contact angles before and after immersion in artificial seawater were used to test the changes in surface wettability. As shown in Fig. 7, the contact angles of ABPs after 12 days of immersion in artificial seawater were all less than the original contact angles. The reason for this phenomenon maybe that the diphenylpyridine side group reacted with sodium in weak alkaline seawater and then hydrolysed in artificial seawater; therefore, the hydrophobic side groups were replaced by hydrophilic sodium carboxylate moieties. The change from hydrophobic groups to hydrophilic groups led to lowered contact angles after immersion in artificial seawater (Fig. 7).

3.7. Weight loss ratios and acid values of different ABPs

Different amounts of hydrolysable domains can result in polymers having different weight loss ratios. Weight loss measurements

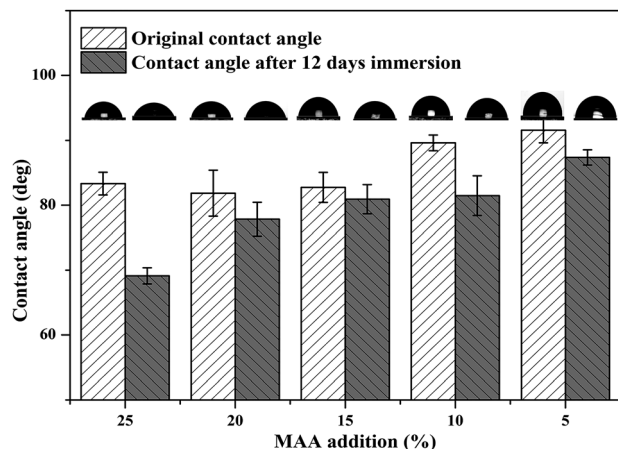


Fig. 7 Original contact angles and the contact angles after 12 days of immersion in artificial seawater of different ABPs. For each surface sample, the reported result is the average of at least five measurements. Error bars represent the standard error.

were carried out to verify whether the amounts of hydrolysable hydrophobic domains were in accordance with the weight loss ratios. As shown in Fig. 8, the weight loss ratios of all ABPs tended to increase over time. Also, it can be observed that ABP-1 exhibited a larger weight ratio than others, whereas ABP-5 had the smallest weight loss ratio. This can be because more MAA species were added during the synthesis of ABP-1, resulting in more labile esters in ABP-1 than in ABP-5 and more hydrolysable domains on the surface. The weight loss ratio trend of the binders was in accordance with the trend of their hydrolysis rates.

Acid value test is another testing method to explain the variation trend of weight loss ratios and hydrolysable domains. As we can see from Table 1, the acid values declined from ABP-1 to ABP-5. The reason for this phenomenon was that there were less MAA species in ABP-5 than in ABP-1 during the synthesis of ABPs. Therefore, there were more carboxyl groups in the molecular structure of AAP-1 than in the molecular structure

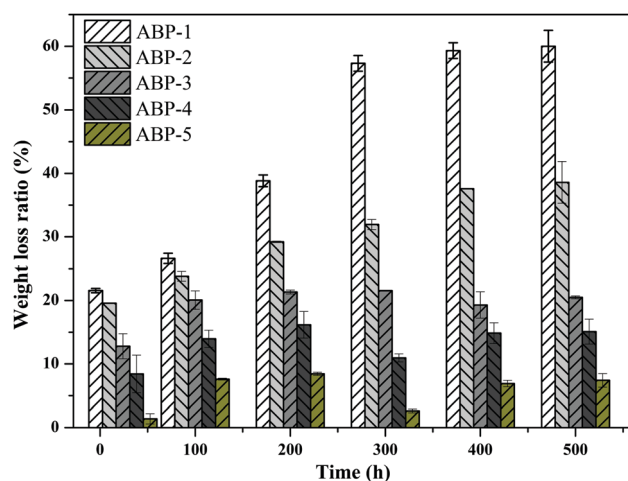


Fig. 8 Weight loss ratios of different ABPs. Error bars represent the standard deviation.

of AAP-5; also, more carboxyl groups led to more hydrolysable domains and more KOH was consumed in the titration process, which led to a decrease in the acid values from ABP-1 to ABP-5.

3.8. Analysis of surface topography and chemical elements

As is known, both composition and surface topography can affect the antifouling performance. Thus, the surface topography of ABPs was observed through a scanning electron microscope. SEM images showed that there were many subuliform protuberances on the surfaces of ABPs before hydrolysis, and ABPs with the maximum amount of hydrolysable hydrophobic diphenyl borane pyridine side groups exhibited the most protuberances (Fig. 9).

After hydrolysis, no protuberances could be found on the surfaces of ABPs, and micropores appeared. Hence, we assumed that the protuberances were agglomerated hydrolysable hydrophobic diphenyl borane pyridine side groups. After hydrolysis of ABPs, the hydrolysable hydrophobic diphenyl borane pyridine side groups peeled off from the backbone, and the protuberances disappeared. To verify this hypothesis, energy dispersive spectroscopy (EDS) was used to measure the changes in elemental compositions of polymer surfaces before and after hydrolysis. The results of the EDS test are shown in the table below (Table 3).

The results of EDS showed that the boron content on the surfaces declined apparently after ABPs were immersed in artificial seawater, which may have been caused by the hydrolysis of ABPs. After the hydrolysable hydrophobic diphenyl borane pyridine side groups peeled off from the backbone to the seawater, the newly formed sodium carboxylate chain

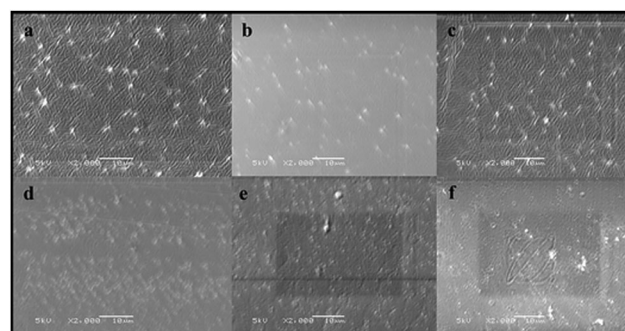


Fig. 9 SEM images of ABPs (a) ABP-1 before hydrolysis, (d) ABP-1 after hydrolysis, (b) ABP-3 before hydrolysis, (e) ABP-3 after hydrolysis, (c) ABP-5 before hydrolysis, (f) ABP-5 after hydrolysis.

Table 3 Chemical compositions of polymer surfaces before and after hydrolysis

Sample	B ^a		O		C	
	Before	After	Before	After	Before	After
ABP-1	20.70	11.84	9.27	19.31	70.03	68.85
ABP-2	19.03	9.17	8.62	23.45	72.35	67.38
ABP-3	18.62	8.76	7.52	23.01	73.86	68.23
ABP-4	17.60	5.55	5.67	29.32	76.73	65.13
ABP-5	16.62	4.91	4.98	30.79	78.40	64.30

^a Feed weight ratio.

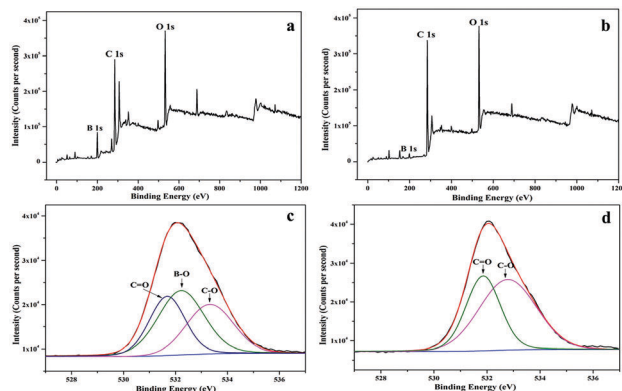


Fig. 10 Survey curves of (a) pristine ABP-1 and its high resolution curves of O 1s (c), (b) ABP-1 immersed in artificial sea water and its high resolution curves of O 1s (d).

segments migrated to the polymer surface. This whole reaction resulted in decreased boron content.

To obtain more information about these hydrolysable surfaces, XPS analysis was used to determine the chemical composition and the nature of the chemical bonding on the surfaces.

Because the peaks of all ABPs were almost the same, ABP-1 with the most distinct boron peaks was chosen to test the chemical compositions. Fig. 10 shows the X-ray photoelectron spectra (XPS) of pristine ABP-1, ABP-1 immersed in artificial seawater, and their high resolution spectra of O 1s.

The binding energies obtained in the XPS experiment were corrected by referencing the C 1s peak to 284.6 eV. In the XPS wide-scan spectrum of pristine ABP-1, the B 1s signal with the respective binding energy (BE) of approximately 193.1 eV is associated with boron in hydrolysable hydrophobic diphenyl borane pyridine side groups, which indicates the successful copolymerization of ABPs.³⁰ After immersion in artificial seawater, the B 1s signal declines sharply (Fig. 10b), which demonstrates that ABP-1 hydrolyses in seawater and the hydrolysable hydrophobic diphenyl borane pyridine side groups peel off from the backbone to seawater. The O 1s core-level spectrum of pristine ABP-1 can be fitted into three components: these three peaks are ascribed to C–O species at 532.7 eV, C=O species at 531.2 eV and B–O species at 532.5 eV.^{31,32} The O 1s core-level spectrum of ABP-1 immersed in seawater can be fitted into two components with BEs at approximately 531.2 eV and 532.7 eV corresponding to C=O and C–O species. Additionally, on the immersed polymer surface, B–O species are not detected at 532.5 eV. These results agree with the EDS data and demonstrate decrease in boron on the surface after immersion in artificial seawater. All these results demonstrate that the antifouling performance of ABPs is achieved by hydrolysis of hydrophobic diphenyl borane pyridine side groups, which endows ABPs with self-polishing properties.

4. Conclusions

A new class of non-metal antifouling ABPs composed of methyl methacrylate, 2-ethylhexyl acrylate, methacrylic acid, and

pyridine-triphenylborane has been successfully developed by the combination of antifouling and environmentally friendly properties. The chemical structures and molecular weights were determined by FTIR and GPC, respectively. All the ABPs exhibited good film-forming and erosion properties in artificial seawater, making them suitable as novel polymeric binders for self-polishing coatings. The gradual hydrolysis of polymer surfaces and hydrolysis products could effectively slow down the colonization of diatoms. The environmentally friendly self-generating idea has been conceptually proven in the current study. In addition, the paints made of ABPs showed excellent antifouling performances in natural seawater. The results from the diatom inhibition test showed that ABPs exhibited negligible influence on the diatom growth rate, which demonstrated that non-metal ABPs were harmless. In addition, the antifouling mechanism of ABPs revealed by EDS and XPS showed that the antifouling performance of ABPs was achieved by the hydrolysis of hydrophobic diphenyl borane pyridine side groups, which endowed ABPs with self-polishing properties.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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